

Robotic CyberKnife radiosurgery for small choroidal melanomas

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Plaque brachytherapy is the most common procedure for the treatment of small choroidal melanoma, especially in posteriorly located tumors. However, there is only little information on outcome after treatment with stereotactic radiosurgery, for example, CyberKnife radiosurgery. We reviewed patients with choroidal melanoma (maximum tumor height 4 mm) treated with CyberKnife radiosurgery. Demographic information, tumor dimension, complications, metastasis and overall survival during the whole follow-up were tracked and analyzed with a specific focus on local tumor control and potential risk factors. One hundred eighty-eight patients (102 female, 54.2%) with a median age of 63 years [interquartile range (IQR): 54–73 years] were analyzed over a median of 46 months (IQR: 24–62 months). Metastasis occurred in 14 patients (7.4%) in median of 18 months after treatment (IQR: 13–47 months) and survival was achieved in 178 patients (94.7%). Within the observation period, eye retention was observed in 166 patients (88.3%). Superior local control was achieved in patients who received a prescription dose of at least 21 Gy or more (91.6%; $P = 0.04$). Other potential risk factors incrementing local control were treatment

planning with missing MRI in advance and too small target delineation during the planning process. Radiosurgery (CyberKnife) in a single, outpatient procedure is suitable for the treatment of small choroidal melanoma. We found local control rate after 3 years compared to the standard treatment with plaque brachytherapy. On the study side, 15 years of CyberKnife treatment allowed to identify risk factors that might increment local control and thus improve treatment regimens. *Melanoma Res* 32: 192–199 Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.

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Introduction

Several studies deal with the survival rates of patients suffering from choroidal melanoma. Smaller tumors tend to have less metastases and therefore a better survival rate [1]. Studies tried to identify factors that influence tumor growth and the development of metastases. Shields *et al.* [1], for instance, conducted a retrospective study that examined 1329 patients with small choroidal melanocytic lesions and tried to determine risk factors for tumor growth. They found that lesions with a tumor thickness of >2 mm and tumors that were located at the posterior pole or the margin of the optic disc showed a higher risk of tumor growth. McLean *et al.* [2] examined 16 factors that could potentially predict the development of metastases. They included 217 patients with small melanomas (tumor volume <1400/cubic mm) and identified four variables that were the best predictive parameters in combination: cell type, largest diameter, scleral extension and mitotic activity.

Thus, many clinicians suppose that the best approach to manage small melanocytic choroidal melanomas is to first

observe the tumor and suspend treatment until there is significant tumor growth [1,3,4]. Yet, a significant part of the patients observed, for instance, by Gass [3] suffered from tumor growth, that made enucleation necessary for about one-third of the patients. However, a treatment suspension of 6 months despite tumor growths did not significantly increase the melanoma-specific mortality [5]. Nevertheless, the study group pointed out that delaying therapy despite tumor growth should not become standard treatment for all patients suffering from posterior uveal melanoma.

After their evaluation of 3806 choroidal nevi, Shields *et al.* developed a rating system helping an ophthalmologist to evaluate tumor progression based on clinical features. They recommend careful watching and early therapy for melanocytic tumors with a greater number of risk factors, in order to achieve sufficient tumor control [6,7].

Originally, a choroidal melanoma was treated by enucleation. Today, one can preserve the globe by using therapy options such as plaque radiotherapy [7,8], stereotactic

radiosurgery [9,10] or external beam radiation [11]. However, small choroidal melanoma with a prominence of less than 6 mm are usually treated using plaque brachytherapy, which is available as a standard therapy since the 1960s. To our knowledge, there is a lack of literature regarding treatment of especially small choroidal melanoma using stereotactic radiosurgery.

In the present study, patients with small choroidal melanoma (maximum tumor thickness of 4.0 mm according to Shields *et al.* [1,12]) were treated with robotic-assisted CyberKnife radiosurgery. We investigated the response to treatment, rates of local control, and tried to identify specific risk factors that improve local control. We also recorded overall survival and eye retention rate as well as complication occurrence and management.

Methods

Study setting

A retrospective review of all patients that were diagnosed with choroidal melanoma and were treated with robotic-assisted CyberKnife radiosurgery between September 2006 and December 2018 at the Department of Ophthalmology of the Ludwig Maximilian University in Munich, Germany, was conducted. The minimum follow-up time was 6 months. In order to focus on small melanoma, only patients with a tumor thickness of 4 mm or less were included in the present study.

A database including age, sex, best-corrected visual acuity (BCVA) at first presentation and each follow-up was built. Upon first presentation, each patient received a full ophthalmologic examination, including assessment of BCVA, slit-lamp biomicroscopy, funduscopy, standardized A-/B-scan ultrasound and photodocumentation. A complete oncologic workup was initiated at the time of diagnosis, if not completed by then.

Patients were followed-up every 3 months after treatment for the first year, every 6 months up to year 5, and annually thereafter. Metastatic disease was excluded at baseline visit by systemic staging [computed tomography (CT) scan] or hepatic ultrasonography, followed by surveillance using hepatic ultrasonography and blood tests every 6 months for the first 5 years and annually thereafter [13]. Local recurrence was determined by an increase in tumor height on ultrasound that was confirmed at two subsequent examinations or an increase of the tumor basal diameter after treatment in comparison with previous photographs at any time after treatment.

Furthermore, complications like increase of intraocular pressure, neovascular glaucoma, vitreous bleeding, retinal detachment were documented. According to the Finger and Kurli staging system [14] patients with radiation retinopathy were categorized in four groups. Written informed consent was obtained before treatment and risks and prognosis as well as treatment options

were discussed with the patient. This study adhered to the tenets of the Declaration of Helsinki and obtained Institutional Review Board approval.

CyberKnife radiotherapy was performed as a standardized outpatient procedure as described previously [14]. In brief, standard retrobulbar anesthesia was performed to achieve akinesia of the globe within the orbit. Axial T2 and gadolinium-contrast-enhanced T1 MRI of the orbit were acquired, followed by a CT scan of the patient's head with 1.0 mm axial slices for planning. CT and MRI scans were overlaid for target volume definition by both ophthalmologists and radiation oncologists. Clinical data such as ultrasound or fundus images were also incorporated into treatment planning. A 1-mm margin in all directions and 2 mm posteriorly was added to the visible tumor to create a planning target volume intended to compensate for potential slight posterior shift of the eyeball during treatment due to resorption of the anesthetic. A non-isocentric inverse algorithm was used in cooperation with a medical physicist to identify optimal beam directions and weights (Multiplan and Precision treatment planning software; Accuray Incorporated, Sunnyvale, California, USA).

Radiation was delivered in a single fraction with a CyberKnife system (G4: 2005–2012; M6: 2013–2018; Accuray Incorporated). The system comprises a 6 MV linear (LINAC) mounted to a 6-axis robotic manipulator and a stereoscopic X-ray imaging system. For treatment, the patient was positioned on a robotic treatment couch, a mask retained the face to prevent motion. The most recent position information was utilized to actively guide the treatment beam to the current tumor location. Radiation delivery was completed within 3 hours from retrobulbar anesthesia to ensure immobility of the eye during the whole treatment.

Study population

Overall, 697 patients treated with robotic-assisted radiosurgery following CyberKnife between January 2005 and August 2019 were reviewed. 188 of these patients were included in this study as they matched with the inclusion criteria. According to Shields *et al.*, main inclusion criterion was small choroidal melanoma [1,12] with tumor height of 4 mm or less. Patients with incomplete data and follow-up time of less than 6 months after treatment were excluded.

Statistical analysis

Overall survival, local control and eye retention rate were calculated applying the Kaplan–Meier product-limit-method implemented in Python (version 3.6.10) using the lifelines package (version 0.25.5). Chi-square test was used to test differences between the groups and descriptive statistics were calculated using SPSS Statistics (version 27; IBM Corp., Armonk, New York, USA). Results

were considered significant if the *P*-value of a logrank test was below 0.05.

Results

Overall, 188 patients were included in the analysis, of them were female (54.3%) and 86 male (45.7%). The median age was 63 years [interquartile range (IQR): 54–73 years]. These patients were treated between September 2006 and December 2018. The median time between CyberKnife treatment and last follow-up visit was 46 months (IQR: 24–62 months). 161 patients (85.6%) were treated with a prescription dose of 21 Gy, 22 patients (11.7%) were treated with lower prescription doses (18–20 Gy) and five patients (2.7%) with prescription doses of 22 Gy. See Table 1 for more detailed patient characteristics.

Local control and eye retention; overall survival and development of metastases

Local tumor control within the observation period was achieved in 165 out of 188 patients (87.8%) after single-session radiosurgery (see Fig. 1 for survival within the first 6 years). The median time between treatment and event of local recurrence was 49 months (IQR: 18–58 months). Local control in patients treated with a minimum of 21 Gy was better than in patients that received less than 21 Gy prescription dose (91.6% vs. 59.1%, *P* = 0.04), respectively. See Fig. 2 for details on the survival of patients treated with the respective prescription dose.

Twenty-three patients (12.2%) developed tumor recurrence during the observation period. Different treatment regimens were applied: 9 patients (39.1%) received a second robotic-assisted radiosurgery (salvage CyberKnife treatment), 13 patients (56.5%) had an enucleation, and a

transscleral exoresection of the tumor was performed in one patient (4.3%, Table 2).

In cases of recurrence, retrospective visual inspection of treatment plan and with current state of knowledge, target delineation and choice of margin was assessed to be too small in all 23 cases. Furthermore, 12 patients (52.2%) with recurrence did not have a contrast MRI in advance (Table 2).

Eye retention was accomplished in 166 (88.3%) of 188 patients. Figure 3 shows the overall eye retention rate during follow-up. 13 patients (6.9%) were enucleated due to local recurrence and eight patients (4.3%) due to neovascular glaucoma with painful loss of vision. One patient (0.5%) specifically wished enucleation because personal reasons.

During the follow-up period, patients receiving radiation doses of 21 Gy or more showed superior eye retention rate (152 patients, 91.6%) in contrast to patients treated with less than 21 Gy (14 patients, 63.6%, respectively, *P* = 0.27).

In total, 10 patients (5.3%) developed liver and four patients (2.1%) combined liver and lung metastases. The median time point was 18 months after treatment (IQR: 14–46 months). Three patients (1.6%) developed metastases during the first year, 9 patients (4.8%) during the second to fifth year and two patients (1.1%) after more than 5 years posttreatment. In total, 10 patients (5.3%) died during the follow-up period, of whom six patients developed metastases.

Tumor thickness, largest basal diameter and TNM classification (eighth AJCC)

Mean tumor thickness at first presentation and last follow-up was 2.9 mm (range: 0.7–4.0 mm, SD = 0.72 mm) and 2.3 mm (range: 0.0–16.00 mm; SD = 1.70), respectively. It decreased significantly (*P* < 0.001). Largest basal diameter (LBD) significantly decreased from 9.0 mm (mean; range: 0.5–15.9 mm; SD = 2.60) at presentation to 8.5 mm (mean; range 0.0–21.7 mm; SD = 3.34) at last follow-up (*P* < 0.001). According to eighth edition of AJCC Cancer Staging, 121 patients (64.4%) were classified as T1 and 67 patients (35.6%) as T2. Moreover, three patients with T1 (1.6%) and 11 patients with T2 (5.9%) staged tumors developed metastases. At last follow-up, 150 tumors were classified as T1 (79.8%), 33 as T2 (17.6%), two as T3 (1.1%) and three as T4 (1.6%), respectively.

Treatment-related complications

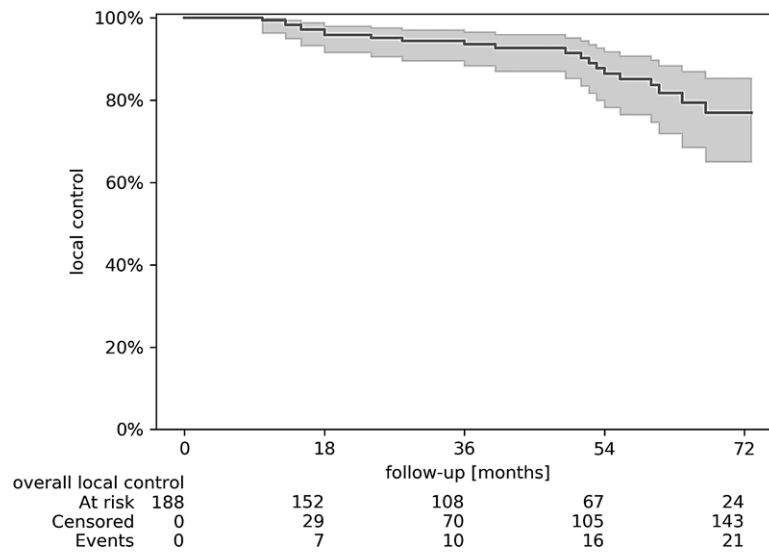
During the study period, 21 patients (11.2%) developed vitreous hemorrhage. Of 42 patients (22.3%) who suffered from serious retinal detachment, 21 needed a pars-plana-vitreotomy for treatment.

Thirty-three patients (17.6%) developed a radiation retinopathy. Only one patient (0.03%) suffered from stage

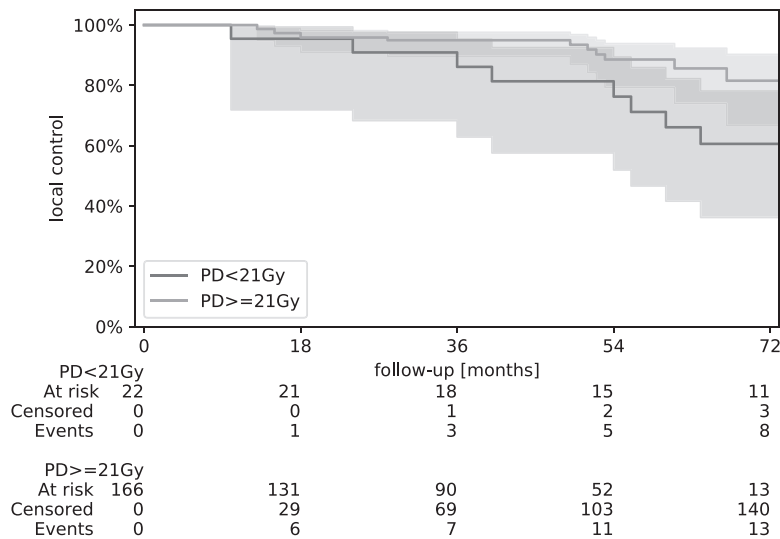
Table 1. Characteristics of patients treated with CyberKnife radiosurgery

Total patients	188
Sex % (male/female)	M/F = 86/102 (M = 45.7%)
Treatment site % (right/left)	RE/LE = 88/100 (R = 46.8%)
Age (years), median (IQR)	63 (54–73 years)
Tumor height (mm), mean ± SD (range)	2.90 ± 0.72 mm (0.70–4.00 mm)
Largest basal diameter (mm), mean ± SD (range)	8.75 ± 2.60 mm (0.50–16.40 mm)
Radiation dose (Gy)	
18 Gys	2
19 Gys	1
20 Gys	19
21 Gys	161
22 Gys	5
Follow-up (months), median (IQR)	46 (42–62 months)
TNM stage as defined by AJCC classification scheme (eighth edition)	
T1	121
T1bN0M0	118
T1bN0M1	3
T2	67
T2bN0M0	56
T2bN0M1	11

LE, left eye; RE, right eye.

Fig. 1.

Overall local control rate in percent (%) during follow-up time in months up to 6 years after CyberKnife treatment.

Fig. 2.

Local control rate in percent (%) subdivided into patients who were treated with less than 21 Gy and patients with a minimum of 21 Gy of radiation dose over the follow-up time in months. Patients who were treated with 21 Gy or more show superior local control rate (in orange).

1 radiation retinopathy without macular involvement, whereas four patients (0.12%), 22 patients (66.67%) and six patients (18.2%) were diagnosed with stage 2, stage 3 and stage 4 radiation retinopathy, respectively, according to the classification of Finger and Kurli [15]. Twenty-two of 188 patients (11.7%) developed neovascular glaucoma within a median time of 31 months (IQR: 14–60 months). Treatment regimen was conservative with drop application only in two patients (13.6%), while 12 patients

(54.5%) needed glaucoma surgery and eight patients (36.4%) were enucleated due to painful neovascular glaucoma.

Discussion

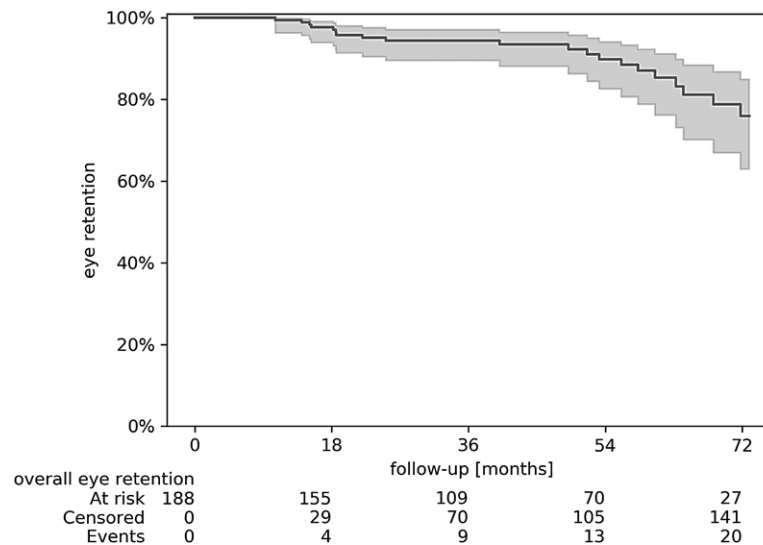
We report on 188 patients with small choroidal melanoma treated with robotic-assisted radiosurgery as a first-line therapy. Mean apical height and mean LBD in these patients were 2.9 mm (SD = 0.72 mm) and 9.0 mm (SD

Table 2 Cases of local recurrence with initial treatment parameters and recurrence treatment regimen

	TNM	Time to recurrence (months)	Radiation dose (Gy)	MRI with CM for treatment planning	Insufficient treatment plan margin	Tumor position (p, pe, a, cb) and growth (EOG)	Management	Metastasis
Case 1	1	88	21	1	1	a, cb	R	No
Case 2	1	36	20	0	1	p	Re-CK	No
Case 3	1	51	21	0	1	pe	Re-CK	No
Case 4	1	15	21	0	1	a, cb	E	No
Case 5	1	53	21	1	1	p	Re-CK	No
Case 6	1	56	20	1	1	p	E	No
Case 7	1	18	21	1	1	p	E	No
Case 8	1	54	20	0	1	p	Re-CK	No
Case 9	1	49	21	1	1	p	Re-CK	No
Case 10	2	18	21	1	1	p	E	No
Case 11	1	82	20	0	1	pe	E	No
Case 12	1	24	20	1	1	p	Re-CK	No
Case 13	2	15	21	1	1	p	E	Yes
Case 14	2	10	18	0	1	p	E	No
Case 15	2	67	21	0	1	p	E	No
Case 16	2	13	21	1	1	e	E	Yes
Case 17	2	64	20	0	1	p	E	No
Case 18	1	28	22	0	1	P	E	No
Case 19	2	61	21	1	1	e; EOG	Re-CK	No
Case 20	2	60	20	0	1	p	E	No
Case 21	2	40	18	0	1	pe	Re-CK	No
Case 22	2	13	21	1	1	a, cb	Re-CK	Yes
Case 23	2	52	21	0	1	pe	E	Yes

0 = no; 1 = yes.

a, anterior; cb, ciliary body; CM, contrast medium; E, enucleation; EOG, extraocular growth; p, posterior; pe, posteroequatoriale; R, resection; Re-CK, second cyberknife treatment; TNM, tumor/node/metastasis.

Fig. 3.

Overall eye retention rate in percent (%) over the follow-up time in months up to 6 years posttreatment.

= 2.60 mm), respectively. So far, the vast part of the literature reports on stereotactic radiosurgery as a secondary treatment option for patients, in whom plaque therapy was not possible due to location or extension.

Local control

During the observation period, local control was achieved in 165 of 188 patients (87.8%). Failure of local control after globe-preserving therapy is a common complication that increases morbidity to the vision and the eye [16].

Therefore, minimizing the risk of local treatment failure and recurrence is fundamental. Plaque brachytherapy, which is available since the 1960s, is the most common treatment of choroidal melanoma. Yet, its local failure rates are reported to range from 0 [17] to 27% [18] with a weighted average of 9.6% [19]. Over the years, the learning curve allowed to reduce local failure rates of plaque brachytherapy through the idea that sometimes the position of the plaque is not sufficient to allow full coverage of the tumor and its margins. In about 36% of the treated

eyes, the iodine-125 plaque needed re-surgery due to plaque displacement [20].

Since we only treated patients with small choroidal melanoma, we are specifically interested in comparing our results to other reports in the literature, mainly treating small choroidal melanoma: Recently, Yipari *et al.* [21] reported on 161 patients treated with iodine-125 episcleral brachytherapy for small choroidal melanoma (apical height 1.0–2.5 mm; LBD < 16.0 mm). The mean follow-up of all patients was 40.7 months. They report a 3-year event-free rate of 99.3%. Only one event of enucleation due to local recurrence was performed within 3 years [21].

Verschuere *et al.* [22] treated 425 patients for small to intermediate-sized choroidal melanomas using Ruthenium-106 plaque brachytherapy with or without transpupillary thermotherapy. The median tumor height was 4.2 mm (range: 1.2–9.3 mm). After 5 years, local control was achieved in 409 patients (96.2%). Including all treated patients in our cohort, we can report a local control rate of 89.7% ($n = 96$ patients) 3 years after CyberKnife treatment.

In Sweden, Bergman *et al.* [23] conducted a large study and observed 579 patients treated with Ruthenium brachytherapy for choroidal melanomas between 1979 and 2003. They did not specify the control rate within their cohort, however, 80 patients (13.8%) were enucleated because of local control failure during the follow-up time of 82 months (median; range: 6–292 months). Still, only 60 patients (10.5%) were diagnosed with small choroidal melanoma.

Reports on local control following stereotactic radiosurgery are scarce, since the treatment has been developed more recently [24–27]. Yet, in the literature, local tumor control following Gamma Knife radiosurgery is reported to range from 84 to 94% 3–8 years posttreatment [10,27–29]. However, to our knowledge, there is only one brief case report about successful treatment of small melanoma using Gamma Knife therapy [9]. After all, studies reporting outcome following Gamma Knife or CyberKnife for the treatment of choroidal melanoma include more patients with medium to large-sized melanomas. The sample size of patients with small melanomas in these studies is often insignificant.

Fakiris *et al.* [10] treated 19 patients following stereotactic radiosurgery and reported a local tumor control rate of 94% 3 and 5 years after treatment (median height: 4.4 mm; median basal diameter: 9 mm). Beyond, Langmann *et al.* [28] treated 60 patients for choroidal melanoma with a median height of 6.7 mm and LBD of 12.2 mm; they reported local recurrence in four cases (6.7%) and therefore had an overall local control rate of 93.3% (follow-up 16–94 months).

In a larger study, Dunavoelgyi *et al.* [25] reported a local tumor control rate of 95.9% 5 years after stereotactic

photon radiotherapy. Altogether 212 patients were treated, yet, only 17 of their patients (8.0%) presented with small melanomas. Thus, direct comparison of the above local control rates with our present study population with a median tumor height of 2.9 mm (range: 0.70–4.00 mm) is not possible.

After several years of treatment with plaque brachytherapy, experience allowed to increase local control in patients treated with plaque brachytherapy through precise plaque position [20]. For our patient population, we were able to identify possible risk factors that might endanger local control following CyberKnife radiosurgery. Hence, we specifically evaluated planning procedures and treatment regimen as influence factors of local recurrence in our study population.

We previously found local control rates for ciliary body melanomas treated with 18–22 Gy at an 70% isodose to be 85% after 3 years [30]. Interestingly, there was a significant difference between patients treated with a prescription dose of 21 Gy (local control rate 90.5%) and patients treated with prescription doses of 20 Gy or less (local control rate 71%) [30].

After all, in our study population 23 of 188 patients (12.2%) developed recurrences during the follow-up period, in line with other study groups [24,25], these were defined as a repetitive measurement of higher tumor prominence or increased tumor basal diameter after treatment in comparison with previous photographs at any time after treatment. We identified insufficient treatment margin (e.g. too small treatment margins, that were delineated and drawn by hand during the beginning of the study period), missing MRI and an applied radiation dose of less than 21 Gy as risk factors for local control failure. Nine of 22 (40.9%) patients that developed a recurrence were treated with a radiation dose of less than 21 Gy. Twelve patients (54.5%) did not receive contrast MRI before treatment, which is essential to distinguish malign from healthy tissue.

Local control rate within 4 years after initial treatment of small choroidal melanoma in our study population was achieved in 76.5% of patients with a radiation dose of less than 21 Gy and in 89.2% of patients with a minimum radiation dose of 21 Gy, respectively ($P < 0.001$, Fig. 2). Yet, it has to be considered that only 22 (11.7%) patients received a prescription dose of less than 21 Gy and thus, these results have to be interpreted with caution.

Metastasis, overall survival and complications

Early detection and immediate treatment are important when growth of a choroidal lesion has been noted. Increasing melanoma thickness is correlated with an increased risk for metastasis [31]. According to Diener-West *et al.* [32], the combined estimate of 5-year mortality of enucleated patients is based on tumor size and was 16% for small melanomas (0–3 mm thickness) [32].

In 2009, Shields *et al.* [31] reported a 10-year risk for melanoma-related metastasis of 12% in small melanomas treated with plaque radiotherapy.

In 2018, the same research group reported a risk of 0.2% for metastasis in 1780 patients with median of 2.6 mm thick tumors in the first, 4.5% in 5 and 8.8% in 10 years [8]. However, they noticed a greater relative risk for the development of metastasis in the thinnest tumors, meaning that flat tumors seem to be more aggressive when it comes to developing metastasis. We report metastasis development in three patients (1.6%) in the first and nine patients (4.8%) within 5 years after treatment. Ten patients (5.3%) died within 5 year follow-up period, of whom six patients died due to metastasis. Our numbers compare well to Shields *et al.* [8] who performed brachytherapy of small choroidal melanomas (mean thickness: 2.6 mm, range: 0.2–3.4 mm). Nevertheless, Yipari *et al.* [21], who treated patients with small melanomas following iodine-125 plaque brachytherapy, noted no metastasis in all included patients. Eight patients (5.0%) died without relation to choroidal melanoma. Dunavoelgy *et al.* [25] as well as Fakiris *et al.* [10] reported higher metastasis rates (15.1 and 31.6%, respectively) following photon radiotherapy and Gamma Knife therapy. However, they treated mainly medium-sized melanomas.

Complication rates following CyberKnife radiosurgery (Table 3) are hard to compare to other study groups, since either treatment regimen or patient population was different [22,23,33]. Interestingly, Yipari *et al.* did not mention any complication or occurrence of enucleation in their study group [21].

In our patient population, vitreous bleeding and retinal detachment were the most frequent complications (11.2 and 22.3%; respectively). These numbers are comparable to Verschuere *et al.* [22] who describe 40 patients (17.1%) with vitreous hemorrhage or retinal detachment with no specific sub differentiation.

We registered the occurrence of neovascular glaucoma in 22 patients (11.7%) in a mean of 37.1 months (range: 1–84 months) post-radiation. Verschuere *et al.* [22] who treated choroidal melanomas with a median height of 4.2 mm with Plaque Brachytherapy reported radiation-induced complications (maculopathy and retinopathy) in 66 patients (15.5%). According to Langmann *et al.* [28] and Yazici *et al.* [33], neovascular glaucoma incidence is associated with higher radiation doses and the proximity of the tumor to the ciliary body. Radiation retinopathy occurred in 33 patients (17.6%) in our patient group; this is comparable to Langmann *et al.* [28] and Yazici *et al.* [33] with 35 and 42%, respectively.

Some limitations need to be considered: first, due to its retrospective character, this study lacks some on patients and the median follow-up time is only 46 months. This

Table 3. Complications in the treatment of choroidal melanoma (0–4 mm thickness) following CyberKnife radiosurgery

Neovascular glaucoma (% of all patients)	22 patients (11.7%)
Medical treatment	2 patients (9.0%)
Surgical treatment	12 patients (54.5%)
Enucleation	8 patients (36.4%)
Vitreous hemorrhage	21 patients (11.2%)
Serous retinal detachment (% of all patients)	42 patients (22.3%)
Pars-plana-vitrectomy (% of patients)	14 patients (33.3%)
Radiation retinopathy (% of all patients)	33 patients (17.6%)

may be regarded as rather short for reports of outcome and complications following radiation treatment of uveal melanoma, but only two patients were followed up for less than 12 months after radiosurgery. Still, most of the previously mentioned studies included data on local control within a follow-up of 5 years, which is comparable to our follow-up time. Since there is a specific gap in literature on outcome following CyberKnife treatment for small melanomas, we consider our results valuable. Second, due to the small number of small melanomas described in other publications, the comparability of our results with theirs and the comparability of treatment by different modalities is impaired.

Summing up, we report on the outcome of patients treated with robotic-assisted radiosurgery (CyberKnife) for small choroidal melanomas with a tumor height of less than 4 mm. A radiation dose of more than 20 Gy was superior regarding outcome and local control. As mentioned by Shields *et al.* [8], flat tumors with thin verges tend to be more aggressive, and therefore harder to capture in the planning process for any kind of radiation therapy. We found that the proper planning of treatment includes contrast-based MRI and a sufficient safety margin around the tumor's edges to increase local control.

Acknowledgements

Conflicts of interest

A.M. and C.F. have previously received speaker's fees from Accuray, Inc. For the remaining authors, there are no conflicts of interest.

References

- Shields CL, Shields JA, Kiratli H, De Potter P, Cater JR. Risk factors for growth and metastasis of small choroidal melanocytic lesions. *Trans Am Ophthalmol Soc* 1995; **93**:259–75; discussion 275.
- McLean MJ, Foster WD, Zimmerman LE. Prognostic factors in small malignant melanomas of choroid and ciliary body. *Arch Ophthalmol* 1977; **95**:48–58.
- Gass JD. Observation of suspected choroidal and ciliary body melanomas for evidence of growth prior to enucleation. *Ophthalmology* 1980; **87**:523–528.
- Davidorf FH, Lang JR. Small malignant melanomas of the choroid. *Am J Ophthalmol* 1974; **78**:788–793.
- Augsburger JJ, Vrabec TR. Impact of delayed treatment in growing posterior uveal melanomas. *Arch Ophthalmol* 1993; **111**:1382–1386.
- Shields CL, Shields JA. Clinical features of small choroidal melanoma. *Curr Opin Ophthalmol* 2002; **13**:135–141.
- Shields CL, Lim LS, Dalvin LA, Shields JA. Small choroidal melanoma: detection with multimodal imaging and management with plaque radiotherapy or AU-011 nanoparticle therapy. *Curr Opin Ophthalmol* 2019; **30**:206–214.

- 8 Shields CL, Sioufi K, Srinivasan A, Di Nicola M, Masoomian B, Barna LE, *et al.* Visual outcome and millimeter incremental risk of metastasis in 1780 patients with small choroidal melanoma managed by plaque radiotherapy. *JAMA Ophthalmol* 2018; **136**:1325–1333.
- 9 Duker JS, Duker JS. Diffuse peripapillary choroidal melanoma that evolved from a small presumed choroidal nevus successfully treated with gamma knife radiotherapy. *Retin Cases Brief Rep* 2016; **10**:338–340.
- 10 Fakiris AJ, Lo SS, Henderson MA, Witt TC, Worth RM, Danis RP, *et al.* Gamma-knife-based stereotactic radiosurgery for uveal melanoma. *Stereotact Funct Neurosurg* 2007; **85**:106–112.
- 11 Malclès A, Kivelä T, Svetlosakova Z, Jean-Louis B, Nguyen AM, Sallit R, *et al.* Small metastasizing choroidal melanomas. *Acta Ophthalmol* 2015; **93**:e160–e166.
- 12 Shields CL, Dalvin LA, Ancona-Lezama D, Yu MD, Di Nicola M, Williams BK Jr, *et al.* Choroidal nevus imaging features in 3,806 cases and risk factors for transformation into melanoma in 2,355 cases: the 2020 Taylor R. Smith and Victor T. Curtin Lecture. *Retina* 2019; **39**:1840–1851.
- 13 Choudhary MM, Gupta A, Bena J, Emch T, Singh AD. Hepatic ultrasonography for surveillance in patients with uveal melanoma. *JAMA Ophthalmol* 2016; **134**:174–180.
- 14 Muacevic A, Nentwich M, Wowra B, Staerk S, Kampik A, Schaller U. Development of a streamlined, non-invasive robotic radiosurgery method for treatment of uveal melanoma. *Technol Cancer Res Treat* 2008; **7**:369–374.
- 15 Finger PT, Kurli M. Laser photocoagulation for radiation retinopathy after ophthalmic plaque radiation therapy. *Br J Ophthalmol* 2005; **89**:730–738.
- 16 Bechrakis NE, Petousis V, Willerding G, Krause L, Wachtlin J, Stroux A, Foerster MH. Ten-year results of transscleral resection of large uveal melanomas: local tumour control and metastatic rate. *Br J Ophthalmol* 2010; **94**:460–466.
- 17 McCannel TA, Chang MY, Burgess BL. Multi-year follow-up of fine-needle aspiration biopsy in choroidal melanoma. *Ophthalmology* 2012; **119**:606–610.
- 18 Leonard KL, Gagne NL, Mignano JE, Duker JS, Bannon EA, Rivard MJ. A 17-year retrospective study of institutional results for eye plaque brachytherapy of uveal melanoma using (125)I, (103)Pd, and (131)Cs and historical perspective. *Brachytherapy* 2011; **10**:331–339.
- 19 Chang MY, McCannel TA. Local treatment failure after globe-conserving therapy for choroidal melanoma. *Br J Ophthalmol* 2013; **97**:804–811.
- 20 Chang MY, Kamrava M, Demanes DJ, Leu M, Agazaryan N, Lamb J, *et al.* Intraoperative ultrasonography-guided positioning of iodine 125 plaque brachytherapy in the treatment of choroidal melanoma. *Ophthalmology* 2012; **119**:1073–1077.
- 21 Yipari RJ, Bena J, Wilkinson A, Suh J, Singh A. Small choroidal melanoma: outcomes following apical height dose brachytherapy. *Br J Ophthalmol* 2021; **105**:1161–1165.
- 22 Verschueren KM, Creutzberg CL, Schalijs-Delfos NE, Ketelaars M, Klijns FL, Haesekeer BI, *et al.* Long-term outcomes of eye-conserving treatment with Ruthenium(106) brachytherapy for choroidal melanoma. *Radiother Oncol* 2010; **95**:332–338.
- 23 Bergman L, Nilsson B, Lundell G, Lundell M, Seregard S. Ruthenium brachytherapy for uveal melanoma, 1979-2003: survival and functional outcomes in the Swedish population. *Ophthalmology* 2005; **112**:834–840.
- 24 Wackernagel W, Holl E, Tarmann L, Mayer C, Avian A, Schneider M, *et al.* Local tumour control and eye preservation after gamma-knife radiosurgery of choroidal melanomas. *Br J Ophthalmol* 2014; **98**:218–223.
- 25 Dunavoelgyi R, Dieckmann K, Gleiss A, Sacu S, Kircher K, Georgopoulos M, *et al.* Local tumor control, visual acuity, and survival after hypofractionated stereotactic photon radiotherapy of choroidal melanoma in 212 patients treated between 1997 and 2007. *Int J Radiat Oncol Biol Phys* 2011; **81**:199–205.
- 26 Krema H, Somani S, Sahgal A, Xu W, Heydarian M, Payne D, *et al.* Stereotactic radiotherapy for treatment of juxtapapillary choroidal melanoma: 3-year follow-up. *Br J Ophthalmol* 2009; **93**:1172–1176.
- 27 Modorati G, Miserocchi E, Galli L, Picozzi P, Rama P. Gamma knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 2009; **93**:40–44.
- 28 Langmann G, Pendl G, Klaus-Müllner, Papaefthymiou G, Guss H. Gamma knife radiosurgery for uveal melanomas: an 8-year experience. *J Neurosurg* 2000; **93**(Suppl 3):184–188.
- 29 Simonová G, Novotný J Jr, Liscák R, Pilbauer J. Leksell gamma knife treatment of uveal melanoma. *J Neurosurg* 2002; **97**(Suppl 5):635–639.
- 30 Liegl R, Schmelter V, Fuerweger C, Foerster MH, Muacevic A, Priglinger SG, *et al.* Robotic cyberKnife radiosurgery for ciliary body melanoma. *Ophthalmol Retina* 2020; **4**:954–956.
- 31 Shields CL, Furuta M, Thangappan A, Nagori S, Mashayekhi A, Lally DR, *et al.* Metastasis of uveal melanoma millimeter-by-millimeter in 8033 consecutive eyes. *Arch Ophthalmol* 2009; **127**:989–998.
- 32 Diener-West M, Hawkins BS, Markowitz JA, Schachat AP. A review of mortality from choroidal melanoma. II. A meta-analysis of 5-year mortality rates following enucleation, 1966 through 1988. *Arch Ophthalmol* 1992; **110**:245–250.
- 33 Yazici G, Kiratli H, Ozyigit G, Sari SY, Cengiz M, Tarlan B, *et al.* Stereotactic radiosurgery and fractionated stereotactic radiation therapy for the treatment of uveal melanoma. *Int J Radiat Oncol Biol Phys* 2017; **98**:152–158.